arXiv:2605.06477v1 [cs.CV] 7 May 2026

GeoStack: A Framework for Quasi-Abelian Knowledge Composition in VLMs

Pranav Mantini
Department of Computer Science
University of Houston
pmantini@uh.edu

Shishir K. Shah
The University of Oklahoma
School of Computer Science
sshah@ou.edu

Abstract

We address the challenge of knowledge composition in Vision-Language Models (VLMs), where accumulating expertise across multiple domains or tasks typically leads to catastrophic forgetting. We introduce GeoStack (Geometric Stacking), a modular framework that allows independently trained domain experts to be composed into a unified model. By imposing geometric and structural constraints on the adapter manifold, GeoStack ensures the foundational knowledge of the base model is preserved. Furthermore, we mathematically demonstrate a weight-folding property that achieves constant-time inference complexity ($O(1)$), regardless of the number of integrated experts. Experimental results across multi-domain adaptation and class-incremental learning show that GeoStack provides an efficient mechanism for long-term knowledge composition while significantly mitigating catastrophic forgetting. Code is available at https://github.com/QuantitativeImagingLaboratory/GeoStack.

1 Introduction and Motivating Work

Single-task models, such as classifiers, gather knowledge to achieve a domain-specific objective. In contrast, multitask learning [Caruana, 1997] aims to incorporate knowledge from multiple objectives for broader applicability. Similarly, incremental learning aims to expand a model's capabilities on novel data, typically within the same domain. At their core, these problems aim to achieve **knowledge composition**.

Multi-Task Learning [Liu et al., 2016] involves joint training strategies on disparate data distributions and quickly becomes infeasible as the number of tasks increases. Furthermore, this approach increases model complexity and requires addressing complex challenges such as class imbalance and hyperparameter selection. Sequential fine-tuning is a viable alternative, where an existing model is fine-tuned with novel data. However, these models are prone to catastrophic forgetting [Kirkpatrick et al., 2017].

Catastrophic forgetting is a phenomenon where connectionist models, when fine-tuned on new data, fail to retain their original knowledge. Classic approaches to address this include Knowledge Distillation [Hinton et al., 2015, Li and Hoiem, 2017], which regularizes student predictions against a frozen teacher, and Data Replay [Rebuffi et al., 2016], which interleaves exemplary data from previous tasks with data from new tasks to maintain consistency.

Adapter-based methods have emerged as a flexible alternative for multitask learning. In this paradigm, models share a large frozen backbone and train a small set of parameters for domain-specific tasks. Houlsby et al. [2019] has trained single-task adapters for text classification tasks, and Stickland and Murray [2019] have proposed multitask adapters for the Recognizing Textual Entailment dataset that match the performance of fine-tuned models.

Preprint.

Single-task adapters generally do not allow for the sharing of information between tasks. To overcome this, Pfeiffer et al. [2021] proposed a two-stage mechanism: (1) **Knowledge Extraction**, where task-specific representations are encapsulated within adapters, and (2) **Knowledge Composition**, which employs a fusion mechanism to combine expertise across adapters for multitask scenarios. Chaichana et al. [2025] proposed Decom-Renorm-Merge that utilizes Singular Value Decomposition (SVD) and renormalization to fuse independently trained checkpoints into a single multitasking model. **Task Arithmetic** proposed by Ilharco et al. [2023] further simplifies this by representing each task as a **task vector** (the difference between fine-tuned and pre-trained weights), which can be linearly summed to merge multiple capabilities into a single model without additional parameters. Furthermore, benchmarks like VL-Adapter [Sung et al., 2022] demonstrate that weight-sharing mechanisms across vanilla adapters can often match the performance of full fine-tuning with significantly less parameter overhead. Such methods aim to learn expertise independently, without the need for simultaneous data access or prohibitive retraining.

Knowledge Extraction using VLMs: VLM-based adapters, particularly those utilizing Contrastive Language-Image Pre-training (CLIP [Radford et al., 2021]) as a backbone, have proven to be an efficient mechanism for rapid domain adaptation. These generally fall into two paradigms: 1) Prompt-based Tuning: Methods such as CoOp Zhou et al. [2022a] and Co-CoOp [Zhou et al., 2022b] optimize learnable prompt vectors to adapt a model for a targeted domain distribution. 2) Adapter-based Tuning: Techniques like CLIP-Adapter [Gao et al., 2025] and Tip-Adapter [Zhang et al., 2021] introduce lightweight bottleneck layers to refine features for specific domains.

Despite these advancements, knowledge composition mechanisms for VLM adapters remain largely nonexistent. We advocate that an ideal framework for knowledge composition must satisfy the following fundamental principles:

1. *Independent Training:* Adapters must be trainable independently without the need for cross-domain or historical data, and the need for joint-training hyperparameters.
2. *Modularity:* Adapters should be modular, allowing the integration of knowledge without re-training the ensemble.
3. *Order-Invariance:* The composed knowledge should be invariant to the order of integration, thus removing the need for combinatorial optimization.
4. *Foundational Preservation:* Composition must not degrade the model's original capabilities or disrupt the foundational knowledge.
5. *Computational Efficiency:* Architectural complexity should remain constant or, at most, grow linearly with each added task.

Recent studies have enabled a deeper understanding of CLIP's geometric properties, specifically the well-known modality gap [Liang et al., 2022] and the canonical relations [Gupta et al., 2026] observed between the feature distributions of independently trained VLMs. Building on these insights, Bilinear CLIP (BiCLIP) [Mantini and Shah, 2026] proposes domain canonicalization for few-shot adaptation by introducing a learnable geometric transformation matrix W. While zero-shot CLIP computes classification probabilities via the dot product $IT^\top$ between image (I) and text (T) features, BiCLIP optimizes the transformed product $IWT^\top$.

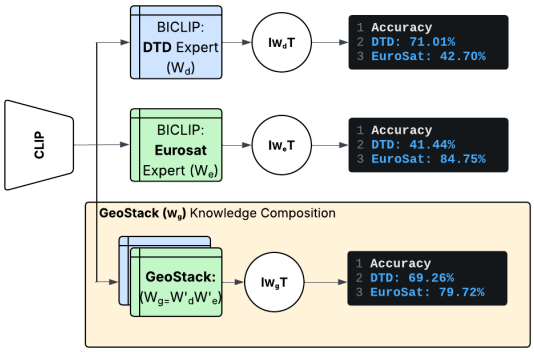

Figure 1: **GeoStack Overview.** Domain-specific adapters (BiCLIP) are not applicable across domains and have reduced generalizability. GeoStack allows domain experts (W'_d, W'_e) to be trained independently and then stacked to enable applicability across multiple tasks.

BiCLIP is an efficient and geometrically interpretable mechanism for domain adaptation. However, these transformations are domain-specific. As shown in Figure 1, a BiCLIP expert optimized for the DTD [Cimpoi et al., 2014] domain achieves 71.01% accuracy on its target but falls to 42.70% on the EuroSAT [Helber et al., 2019] dataset. Conversely, a EuroSAT expert achieves 84.75% on its own domain but drops to 41.44% on DTD.

We propose **Geometric Stacking (GeoStack)**, a modular knowledge composition framework designed to aggregate expertise from multiple domain-specific adapters into a multi-expert model with zero additional inference complexity. Specifically, BiCLIP adapters are trained with geometric constraints to produce domain-specific Geometric Layers (GeoLayers) that can be stacked onto one another for multitask performance. As shown in Figure 1, GeoStack performs knowledge composition to maintain the performance across both DTD (69.26%) and Eurosat (79.72%) datasets. This approach allows an arbitrary number of experts to be composed via matrix multiplication and folded into a single weight matrix, maintaining $O(1)$ complexity regardless of the number of domains.

Our primary contributions are as follows:

- **The GeoStack Framework**: We introduce GeoStack, a modular framework for knowledge composition in VLMs. We derive the geometric constraints necessary to ensure the stability of the framework.
- **Theoretical Foundations of Stackability**: We define the metrics and conditions under which domain-specific experts can be stably composed using GeoStack.
- **The Weight-Folding Property**: We demonstrate that GeoStack adapters enable multitask inference with constant-time complexity ($O(1)$), independent of the number of experts used in the composition.
- **Empirical Validation**: We conduct extensive experiments across multi-domain adaptation and class-incremental learning, demonstrating GeoStack's superior performance and its resistance to catastrophic forgetting.

2 GeoStack Theory

2.1 Preliminaries

CLIP projects images and textual prompts into a shared embedding space $\mathbb{R}^d$, yielding features I and T. In the zero-shot setting, the similarity (dot product) between these representations is used to compute the posterior for classification. Given a matching positive pair (I, T^+) and an unmatched negative pair (I, T^-), the classification decision boundary and the resulting margin M_z are expressed as:

$$IT^{+\top} > IT^{-\top} \implies M_z = IT^{+\top} - IT^{-\top} > 0 \tag{1}$$

However, CLIP is trained on generic web data, and this pre-trained geometric boundary is often inadequate in specialized domains.

BilinearCLIP [Mantini and Shah, 2026] (BiCLIP) hypothesizes that the domain-specific decision boundary can be recovered by applying a geometric transformation $W \in \mathbb{R}^{d\times d}$ to the image features ($I' = IW$). For a domain $\mathcal{D}_a$, the BiCLIP margin M_a is defined as:

$$M_a = I_a W_a T_a^{+\top} - I_a W_a T_a^{-\top} > 0 \tag{2}$$

While W_a optimizes the decision boundary for domain $\mathcal{D}_a$, it degrades the generalization capabilities of CLIP, resulting in a model that is not applicable to other domains.

2.2 GeoStack: Problem Formulation

Building on BiCLIP, GeoStack is a modular architecture that allows the composition of multiple experts via sequential matrix multiplication. The composite operator for domains $\mathcal{D}_a$ and $\mathcal{D}_b$ is defined as $W_g = W_a W_b$. The primary challenge is in ensuring that the subsequent expert W_b does not destroy the margin M_a previously established for $\mathcal{D}_a$. To ensure framework viability, the composite margin M_g must satisfy:

$$I_a W_g T_a^{+\top} - I_a W_g T_a^{-\top} > 0 \quad \text{and} \quad I_b W_g T_b^{+\top} - I_b W_g T_b^{-\top} > 0 \tag{3}$$

For these inequalities to hold, the original margin M_a must remain positive under the influence of subsequent operators. We require the framework to satisfy the stability condition: $M_a > 0 \implies M_g > 0$.

2.3 GeoStack: Geometric Constraints for Multi-Domain Composition

GeoStack ensures this margin stability by imposing two geometric and structural constraints from BiCLIP:

Upper Triangular Closure: We restrict each adapter W to the set of upper-triangular matrices $\mathcal{U}$. Because $\mathcal{U}$ is closed under multiplication, any composed operator $W_{total} = W_a W_b \dots W_n$ remains upper-triangular, ensuring the composite operator is always a valid member of the same transformation class.

Perturbation Prior: Each learnable adapter W is initialized as the identity matrix $\mathbf{I} \in \mathcal{U}$. This initialization acts as a geometric prior. Consequently, the learned transformation can be viewed as a perturbation $W = \mathbf{I} + \Delta$, where $\Delta \in \mathcal{U}$ represents the domain-specific geometric shift.

2.4 Perturbation Minimization Theory

By defining each domain expert as a perturbation $W_i = \mathbf{I} + \Delta_i$, the GeoStack composition of two experts is given by:

$$W_a W_b = (\mathbf{I} + \Delta_a)(\mathbf{I} + \Delta_b) = \mathbf{I} + \Delta_a + \Delta_b + \Delta_a \Delta_b \tag{4}$$

When the learned perturbations Δ_a and Δ_b are small, their product term becomes negligible ($\Delta_a \Delta_b \approx \mathbf{0}$) as a second-order effect. The composed margin for domain $\mathcal{D}_a$ becomes:

$$M_g \approx I_a(\mathbf{I} + \Delta_a + \Delta_b){T_a^+}^\top - I_a(\mathbf{I} + \Delta_a + \Delta_b){T_a^-}^\top \approx M_a + \underbrace{\left[I_a \Delta_b {T_a^+}^\top - I_a \Delta_b {T_a^-}^\top\right]}_{\epsilon} \tag{5}$$

Here, ϵ represents the inter-domain interference caused by Domain $\mathcal{D}_b$. The stability of the composed margin relies inherently on the spectral norms of the perturbations ($\|\Delta_b\|_2 < \delta$) remaining small. This structural guarantee ensures that $\|\epsilon\| \ll M_a$. Therefore, the composed GeoStack margin M_g remains positive as long as M_a is sufficiently discriminative and the **spectral norm of the perturbations is minimized**.

2.5 Properties of GeoStack

The perturbation minimization theory underpinning GeoStack yields highly desirable mathematical properties for knowledge composition.

1. Quasi-Additive Composition: The multiplicative composition of GeoStack effectively reduces to a quasi-additive operation: $W_g \approx \mathbf{I} + \sum_{i=1}^{n} \Delta_i$. This property ensures that as new domains are added, the foundational knowledge of CLIP and previously learned experts is preserved.

2. Quasi-Abelian composition: A direct corollary of the quasi-additive property is the commutativity of the GeoStack. As $W_a W_b \approx \mathbf{I} + \Delta_a + \Delta_b$, the order of composition becomes largely irrelevant ($W_a W_b \approx W_b W_a$). This grants the GeoStack framework a quasi-Abelian property, allowing for greater flexibility in knowledge composition.

3. The Stacking Metric: We utilize the normalized orthogonality error as a proxy for stacking compatibility. Substituting $W = \mathbf{I} + \Delta$, the error expands to $\|WW^\top - \mathbf{I}\|_F^2 \approx \|\Delta + \Delta^\top\|_F^2$ (See Appendix A.1). Since the Frobenius norm upper-bounds the spectral norm ($\|\Delta\|_F \geq \|\Delta\|_2$) and is more computationally efficient to calculate, it serves as a practical upper bound for the interference $\|\epsilon\|$. This yields a practical stacking metric: operators with high orthogonality error violate the condition $\|\epsilon\| < M_a$, subsequently leading to catastrophic forgetting.

4. The Folding Trick ($O(1)$ Inference): GeoStack enables Zero-Overhead Inference via weight folding. In CLIP, the visual projection head is a matrix $P \in \mathbb{R}^{d' \times d}$. Since each adapter W_k is a $d \times d$ matrix, the entire stack can be pre-computed into a single effective projection matrix: $P_{eff} = P \cdot \prod_{k=0}^{n-1} W_k$

This property ensures that the inference complexity is constant ($O(1)$) with respect to the number of tasks. Mathematically, P_{eff} is structurally identical to the original vanilla CLIP projection, meaning

GeoStack provides multi-domain expertise with zero additional latency or memory footprint during deployment.

2.6 Limitations: Margin Erosion in Deep Stacking

Margin Erosion: The Quasi-Additive Property ($W_g \approx \mathbf{I} + \sum \Delta_i$) implies a linear accumulation of error. The total inter-domain interference for domain $\mathcal{D}_a$ is the sum of perturbations from all n subsequent experts: $\epsilon_{total} = \sum_{i \neq a}^{n} \left[I_a \Delta_i T_a^{+\top} - I_a \Delta_i T_a^{-\top} \right]$

As the stack deepens, the stability condition $\|\epsilon_{total}\| < M_a$ is eventually violated—a phenomenon we term **Margin Erosion**. This leads to a gradual degradation in domain-specific performance, eventually regressing to sub-optimal zero-shot CLIP performance or causing a manifold collapse as the error accumulates.

3 GeoLayer

A **Geometric Layer** (GeoLayer) is an evolution of the BiCLIP adapter, optimized with geometric constraints to enable knowledge composition. While a BiCLIP adapter ($W \in \mathbb{R}^{d \times d}$) is optimized with the objective of aligning image and text features within a single domain, it often disrupts foundational knowledge, resulting in catastrophic forgetting. In contrast, a GeoLayer is trained with a dual objective: (1) of achieving domain-specific alignment, while (2) satisfying geometric constraints to preserve previous knowledge. These GeoLayers can be stably composed into a **GeoStack** to enable a multi-domain expert without catastrophic forgetting.

Alignment Objective: We utilize the InfoNCE contrastive loss for domain alignment. Given a batch of N image-text feature pairs (I_j, T_j) from domain $\mathcal{D}_i$, we compute the transformed image features $I'_j = I_j W_i$ and their corresponding text embeddings T_j. The alignment loss is defined as:

$$\mathcal{L}_{align} = -\frac{1}{2N} \sum_{j=1}^{N} \left[\log \frac{e^{\text{sim}(\mathbf{I}'_j, \mathbf{T}j)/\tau}}{\sum_{k=1}^{N} e^{\text{sim}(\mathbf{I}'_j, \mathbf{T}_k)/\tau}} + \log \frac{e^{\text{sim}(\mathbf{I}'_j, \mathbf{T}j)/\tau}}{\sum_{k=1}^{N} e^{\text{sim}(\mathbf{I}'_k, \mathbf{T}_j)/\tau}} \right] \tag{6}$$

where τ is the temperature parameter and $\text{sim}(\cdot)$ denotes cosine similarity. This objective allows the GeoLayer to learn a domain-specific transformation that aligns the image features with their corresponding text modality for classification.

Stackability Objective: To ensure that each GeoLayer satisfies the stability requirements derived in Section 2, we minimize the Frobenius norm of the deviation from orthogonality, which effectively bounds the spectral norm $\|\Delta_i\|_2$. We define the Orthogonality Loss as: $\mathcal{L}_{ortho} = \|W_i^\top W_i - \mathbf{I}\|_F^2$.

This objective ensures that the learned perturbation Δ_i remains minimal. Furthermore, by enforcing W_i to remain in the neighborhood of an orthogonal matrix ensures the transformation is near-isometric. This preserves the feature norms ($\|IW_i\| \approx \|I\|$) during both training and inference.

Convex Orthogonality Alignment Loss: The final optimization objective for a GeoLayer is a convex combination of the alignment and stackability objectives. We define the **Convex Orthogonality Alignment (COA)** Loss as:

$$\mathcal{L}_{COA} = (1 - \lambda)\mathcal{L}_{align} + \lambda \mathcal{L}_{ortho} \tag{7}$$

This formulation enables a calibration of the GeoLayer's behavior. As $\lambda \to 0$, the objective prioritizes domain-specific alignment. Conversely, as $\lambda \to 1$, the objective prioritizes the stability requirement for knowledge composition.

4 Experimental Methodology

We evaluate the efficiency of GeoStack on two Vision problems: Multi-Domain Adaptation (MDA) and Class-Incremental Learning (CIL). The objective is to quantify the performance of GeoStack across disparate knowledge domains and its ability to handle catastrophic forgetting.

Implementation: All experiments were conducted on a NVIDIA GeForce RTX 2080 Ti GPU. We use OpenCLIP's **ViT-B/16** as the backbone encoder, keeping all weights frozen while learning task-

specific GeoLayers. Since each GeoLayer is constrained to an upper-triangular matrix $W \in \mathbb{R}^{d\times d}$, this reduces the learnable parameters by approximately 50% ($\approx 1.3 \times 10^5$ parameters) compared to a full $d \times d$ transformation. Each GeoLayer is trained in isolation, independent of other domains, ensuring constant training complexity regardless of the total number of domains. We utilize the AdamW optimizer with a learning rate of 1×10^{-4} and a batch size of 32, and train for 30–50 epochs. For the COA loss, we fix $\lambda = 0.95$ for all datasets and increase it to 0.99 for domain-specific datasets.

4.1 Multi-Domain Adaptation (MDA)

In this problem setting, we aim to adapt a foundation model to multiple target domains $\{\mathcal{D}_1, \mathcal{D}_2, \ldots, \mathcal{D}_n\}$ simultaneously. Traditional MDA often requires joint training on data from all domains. The modular nature of GeoStack makes it a great candidate for MDA, where GeoLayers can be trained on individual domains separately and then stacked on each other to create a unified multi-domain model. This approach enables the model to perform well across all target domains without requiring simultaneous access to the data or complex joint optimization.

Dataset Categorization: To evaluate GeoStack, we curate a suite of datasets representing diverse semantic complexities. The datasets are categorized as: **General Objects** consisting of ImageNet-1K (i) [Deng et al., 2009] and Caltech-101 (c) [Fei-Fei et al., 2004], **Fine-Grained Objects** consisting of Flowers-102 (f) [Nilsback and Zisserman, 2008] (f) and Food-101 (fo) [Bossard et al., 2014], and **Domain-Specific Images** consisting of EuroSAT (e) [Helber et al., 2019], DTD (d) [Cimpoi et al., 2014]. This selection allows us to quantify the model's capacity to preserve foundational general-object knowledge while simultaneously adapting to fine-grained domains and specialized distributions.

4.1.1 Multi-Domain Knowledge Composition with GeoStack

We adopt a two-stage process for knowledge integration, inspired by AdapterFusion [Pfeiffer et al., 2021]:

Stage 1 - Knowledge Extraction: For each domain $\mathcal{D}_k$, a dedicated GeoLayer is trained in isolation using the COA loss under a 16-shot (16 samples per class) protocol. This stage extracts domain-specific expertise while ensuring it is composable.

Stage 2 - Knowledge Composition: To create a unified multi-expert model, we compose the independent GeoLayers into a single **GeoStack**. For example, a stack sequence denoted as $i \rightarrow c \rightarrow fo \rightarrow e$ computes the final transformation as: $W_g = W_i \cdot W_c \cdot W_{fo} \cdot W_e$. Here, the arrows ($\rightarrow$) denote the stacking order, where the initial GeoLayer for (i) forms the base and subsequent layers are appended to the transformation chain. The geometric constraints enforced during Stage 1 ensure that the resulting product W_g remains stable for all domains.

4.1.2 QuadStack: Results and Discussion

While GeoStack can be composed at any arbitrary depth, our analysis focuses on a **Quad-Stack** (Depth-4) configuration. A dual or triple stack may often fail to reveal the potential for long-term instability. By evaluating a Depth-4 composition, we demonstrate GeoStack's capability to maintain cross-domain performance while effectively thwarting catastrophic forgetting.

To evaluate the stability of GeoStack, we define three stacks of increasing domain complexity: (1) the **Easy Stack** ($i \rightarrow c \rightarrow fo \rightarrow e$), following a coarse-to-fine semantic transition; (2) the **Moderate Stack** ($i \rightarrow fo \rightarrow e \rightarrow d$), representing a progression to increasing complexity; and (3) the **Hard Stack** ($i \rightarrow e \rightarrow d \rightarrow f$) representing a departure from general image domain to domain specific visual content.

We compare GeoStack against Task Arithmetic [Ilharco et al., 2023] (TA) by linearly summing the learned perturbations Δ_i. TA treats knowledge composition as an additive operation ($\mathbf{I} + \alpha \sum_i \Delta_i$). We empirically observe that while α is often tuned in TA to balance task performance and interference, setting α to match the orthogonality error (OE) of GeoStack results in a degradation of performance. Consequently, we set $\alpha = 1$ for our primary baseline comparisons. We compare classification accuracy on five configurations: **1. ZS:** The vanilla zero-shot CLIP model without any adapters, **2. TA (BiCLIP):** Linear task arithmetic using bilinear adapters. **3. TA (Geo):** Linear task arithmetic using GeoLayers (trained with COA loss). **4. BiCLIP:** A naive stacking baseline where each domain

Stack	Dataset	ZS	TA (BiCLIP)	TA (Geo)	BiCLIP [OE]	GeoStack [OE]
Easy Stack $i \to c \to fo \to e$	ImageNet (i)	66.6%	60.2%	66.2%	67.1% [0.022]	**69.3% [0.010]**
	Caltech-101 (c)	90.0%	87.6%	90.0%	92.9% [0.022]	**93.1% [0.010]**
	Food-101 (fo)	88.7%	85.4%	88.0%	88.2% [0.022]	**89.5% [0.010]**
	EuroSAT (e)	47.5%	31.5%	37.8%	**84.9% [0.022]**	84.1% [0.010]
	Average	73.2%	66.2%	70.5%	83.3% [0.022]	**84.0% [0.010]**
Moderate Stack $i \to fo \to e \to d$	ImageNet (i)	66.6%	62.0%	67.9%	57.2% [0.052]	**65.4% [0.012]**
	Food-101 (fo)	88.7%	85.3%	88.5%	82.6% [0.052]	**87.1% [0.012]**
	EuroSAT (e)	47.5%	83.8%	82.6%	83.2% [0.052]	**83.3% [0.012]**
	DTD (d)	42.8%	39.3%	42.4%	**69.7% [0.052]**	66.7% [0.012]
	Average	61.4%	67.6%	70.4%	73.2% [0.052]	**75.6% [0.012]**
Hard Stack $i \to e \to d \to f$	ImageNet (i)	66.6%	49.0%	62.1%	52.6% [0.070]	**62.8% [0.013]**
	EuroSAT (e)	47.5%	82.0%	78.3%	81.7% [0.070]	**82.8% [0.013]**
	DTD (d)	42.8%	67.7%	60.0%	**69.5% [0.070]**	66.1% [0.013]
	Flowers-102 (f)	71.0%	52.3%	60.9%	**86.8% [0.070]**	85.8% [0.013]
	Average	57.0%	62.8%	65.3%	72.6% [0.070]	**74.4% [0.013]**

Table 1: **Quad-Stack Multi-Domain Adaptation results.** We evaluate stackability across three tiers of geometric difficulty, comparing Task Arithmetic (TA) against GeoStack.

expert is trained as a standard bilinear adapter W_i without the orthogonality constraint ($\lambda = 0$), and **5. GeoStack [OE] (Proposed):** Where GeoLayers are trained with the COA loss and then stacked.

The results, summarized in Table 1, demonstrate an intuitive correlation between the geometric complexity of target domains, the accumulation of Orthogonality Error (OE), and the preservation of foundational knowledge. Notably, Task Arithmetic (TA) with the constrained GeoLayers yields significant gains over unconstrained BiCLIP, ImageNet accuracy improves from 49.0% to 62.1% in the Hard Stack. In BiCLIP, the introduction of out-of-distribution domains like EuroSAT (e) and DTD (d) results in a collapse of the foundational knowledge. In the Hard Stack, ImageNet accuracy plummets from 66.6% to 52.6% as the cumulative OE increases to 0.070. This confirms that without geometric constraints, subsequent experts distort the existing knowledge of previous domains. Conversely, GeoStack, attributing to the COA loss, maintains ImageNet accuracy at 62.8% with an OE of 0.013. GeoStack consistently provides superior **Average** classification accuracy by maintaining the foundational knowledge, thus validating that geometric constraints are required for stable, modular multi-domain composition.

Additional results for dual, triple, and hexa-stack are provided in Appendix B.1.

4.2 Class-Incremental Learning (CIL)

In this setting, the objective is to learn a sequence of novel classes without sacrificing the ability to recognize previously learned ones. Catastrophic forgetting remains a primary challenge in CIL, resulting in a model that performs poorly on older tasks.

GeoStack offers a unique architectural solution for knowledge composition in CIL. GeoLayer can be trained for each incremental batch of classes and composed into a stack, while keeping the backbone parameters frozen. This makes CIL an ideal problem for quantifying catastrophic forgetting. GeoLayers can be considered as non-disruptive if the composed margin remains wide enough to incorporate new class features while maintaining the performance on the original classes.

4.2.1 Knowledge composition for Class-Incremental Learning

We conduct experiments on CIFAR-100 [Krizhevsky, 2009], consisting of 100 categories of general objects. In this experiment, we define n disjoint tasks $\{\mathcal{T}_0, \mathcal{T}_1, \ldots, \mathcal{T}_{n-1}\}$, where each task introduces a new set of classes. We set $n = 4$ and partition the dataset into four tasks consisting of 25 disjoint classes. For each task $\mathcal{T}_k$, we train a GeoLayer W_k using a 16-shot protocol independently. The cumulative knowledge of the model after n tasks is represented by the composite operator: $W_g = \prod_{k=0}^{n-1} W_k$.

We quantify the model's capability for Incremental Learning and Knowledge Retention. In the former, we measure the model's ability to recognize all classes seen so far (Global Accuracy); in the latter, we measure the model's ability to retain knowledge from the initial classes as newer tasks are added to the stack (Task-0 Retention).

Task	Classes	ZS	BiCLIP	TA	GeoStack
T0	25	71.88	**86.20**	77.92	77.92
T1	50	68.44	**74.60**	69.96	74.18
T2	75	66.89	63.99	65.76	**70.47**
T3	100	68.11	60.08	61.79	**69.47**

Evaluation	BiCLIP	TA	GeoStack
After T0	**86.20**	77.92	77.92
After T1	81.88	79.48	**77.20**
After T2	76.60	**77.28**	76.92
After T3	72.04	74.00	**75.80**
Decay (Δ)	-14.16	-3.92	**-2.12**

Table 2: **CIFAR-100 Incremental Learning Results.** *Left:* Global accuracy on all seen classes. *Right:* Retention of Task-0 (first 25 classes). Zero-Shot (ZS) baseline for Task-0 is 71.88%.

4.2.2 QuadStack: Results and Discussion

The results in Table 2 (left) demonstrate that GeoStack's ability to facilitate knowledge composition significantly minimizes catastrophic forgetting. While the BiCLIP baseline initially achieves a high 86.20% accuracy on Task 0, this performance degrades rapidly as the stack depth increases. In the incremental learning setting, BiCLIP's accuracy drops by approximately 8.7% per additional task, eventually falling to 60.08%—which is 8% below the Zero-Shot CLIP baseline. Similarly, TA (using GeoLayers) falls below Zero-Shot by T2. This indicates that without geometric constraints, the models are susceptible to catastrophic forgetting.

In contrast, GeoStack maintains an accuracy of 69.47% after composing knowledge from four independently trained tasks, outperforming the baseline by 9.39%. The Retention experiment in Tables 2 (right) further highlights GeoStack's resistance to catastrophic forgetting: while BiCLIP loses 14.16% of its original knowledge, TA loses 3.92%, while GeoStack's geometric constraint limits it to 2.12% (77.92% to 75.80%). These findings are particularly encouraging, as successful knowledge composition from independently trained models remains a complex problem in current literature. We attribute this success to the inherent robustness of CLIP's feature space and the effectiveness of the geometric constraints in preserving the previous knowledge successfully.

4.3 Analysis of GeoStack Properties

In this section, we validate the properties of GeoStack by testing its Abelian nature and performing a deep-stacking stress test to examine the margin erosion.

1. The Abelian Property: The Abelian property (commutativity) is a highly desirable characteristic

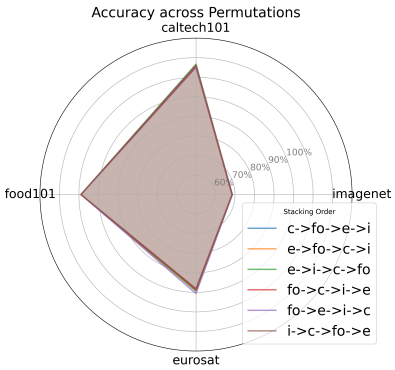

Figure 2: Spider plot showing accuracy invariance across four stacking permutations.

Dataset	Mean (%)	σ	Range
ImageNet	69.29	0.04	0.09
Caltech101	92.97	0.27	0.74
Food101	89.49	0.02	0.07
EuroSAT	84.49	0.42	1.25

Table 3: Accuracy statistics over permutations.

for modular knowledge composition. It allows the GeoLayers to be stacked in any arbitrary order without the need for a combinatorial optimization. To validate this property, we evaluate a QuadStack

configuration consisting of four domain experts from MDA: $\mathcal{D}_{ImgNet}(i)$, $\mathcal{D}_{Caltech}(c)$, $\mathcal{D}_{Food}(fo)$, and $\mathcal{D}_{Euro}(e)$. We measure the performance across multiple stacking permutations to observe sensitivity to ordering.

As illustrated in the spider plot (Fig. 2), the performance for each domain remains consistent regardless of the sequence. For instance, the accuracy on EuroSAT remains stable at $84.49 \pm 0.42\%$, even when its position in the stack is shifted from the base to the final layer. As shown in Table 3, the negligible variance across all tested permutations confirms the Quasi-Abelian nature of GeoStack under the geometric constraints.

2. Margin Erosion and Deep Stacking: To quantify GeoStack's behavior under deep-stacking conditions, we conduct a 10-task class-incremental learning experiment on the CIFAR-100 dataset. The dataset is partitioned into 10 sequential tasks, each containing 10 disjoint classes. We measure the retention of Task-0 accuracy and the ImageNet foundational knowledge to quantify catastrophic forgetting as the stacking depth increases from $n = 1$ to $n = 10$.

As illustrated in Figure 3, GeoStack exhibits robust resistance to catastrophic forgetting; it retains an accuracy of 56.00% on the initial task, significantly outperforming the BiCLIP baseline, which drops to 21.50%. The results indicate that while BiCLIP initially achieves higher peak performance on the first task (88.9%), it quickly leads to an exponential decay with stacking depth. In contrast, GeoStack exhibits a steady and linear degradation in performance. Most notably, GeoStack preserves the model's foundational integrity with a final ImageNet accuracy of 57.2%, representing a 19.4% margin over Task Arithmetic with BiCLIP. This confirms that GeoStack preserves foundational knowledge under deep stacking.

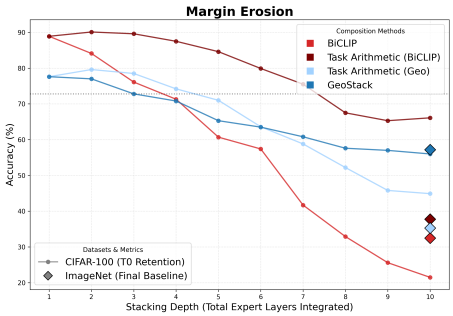

Figure 3: Comparison of Task-0 accuracy and ImageNet knowledge retention across 10 incremental tasks.

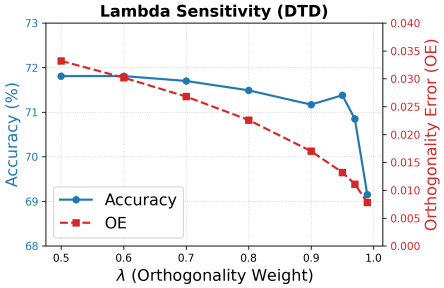

Figure 4: Sensitivity analysis of the geometric constraint weight λ on the DTD dataset.

3. Sensitivity Analysis of λ: We perform a sensitivity analysis on λ, varying it in the range 0.5 to 0.99. Results shown in Fig 4 demonstrate that increasing λ yields an exponential reduction in Orthogonality Error—decreasing from 0.0332 to 0.0078—with a marginal 2.6% impact on accuracy. This confirms that a high λ successfully enables stackability, which is crucial for maintaining foundational integrity.

Further analysis on the Stacking metric is provided in Appendix C.1.

5 Conclusion

In this work, we presented GeoStack, a modular framework that solves the trade-off between knowledge accumulation and catastrophic forgetting in VLM adapters. By enforcing geometric constraints through a novel Convex Orthogonality Alignment (COA) loss, we developed GeoLayers that are stackable for knowledge composition while preserving the foundational CLIP latent space. We demonstrated that an arbitrary number of domain experts can be folded into a single $O(1)$ projection matrix. Experiments on Multi-Domain Adaptation and Class-Incremental Learning reveal that GeoStack matches the performance of domain-specific models while significantly mitigating catastrophic forgetting.

References

L. Bossard, M. Guillaumin, and L. Van Gool. Food-101–mining discriminative components with random forests. In *European conference on computer vision*, pages 446–461. Springer, 2014.

R. Caruana. Multitask learning. *Machine learning*, 28(1):41–75, 1997.

Y. Chaichana, T. Trachu, P. Limkonchotiwat, K. Preechakul, T. Khandhawit, and E. Chuangsuwanich. Decom-renorm-merge: Model merging on the right space improves multitasking, 2025. URL `https://arxiv.org/abs/2505.23117`.

M. Cimpoi, S. Maji, I. Kokkinos, S. Mohamed, and A. Vedaldi. Describing textures in the wild. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 3606–3613, 2014.

J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei. Imagenet: A large-scale hierarchical image database. In *2009 IEEE Conference on Computer Vision and Pattern Recognition*, pages 248–255, 2009. doi: 10.1109/CVPR.2009.5206848.

L. Fei-Fei, R. Fergus, and P. Perona. Learning generative visual models from few training examples: An incremental bayesian approach tested on 101 object categories. In *2004 conference on computer vision and pattern recognition workshop*, pages 178–178. IEEE, 2004.

P. Gao, S. Geng, R. Zhang, T. Ma, R. Fang, Y. Zhang, H. Li, and Y. Qiao. Clip-adapter: Better vision-language models with feature adapters, 2025. URL `https://arxiv.org/abs/2110.04544`.

S. Gupta, S. Kansal, S. Jegelka, P. Isola, and V. Garg. Canonicalizing multimodal contrastive representation learning, 2026. URL `https://arxiv.org/abs/2602.17584`.

P. Helber, B. Bischke, A. Dengel, and D. Borth. Eurosat: A novel dataset and deep learning benchmark for land use and land cover classification. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 12(7):2217–2226, 2019.

G. Hinton, O. Vinyals, and J. Dean. Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*, 2015.

N. Houlsby, A. Giurgiu, S. Jastrzebski, B. Morrone, Q. De Laroussilhe, A. Gesmundo, M. Attariyan, and S. Gelly. Parameter-efficient transfer learning for nlp. In *International conference on machine learning*, pages 2790–2799. PMLR, 2019.

G. Ilharco, M. T. Ribeiro, M. Wortsman, S. Gururangan, L. Schmidt, H. Hajishirzi, and A. Farhadi. Editing models with task arithmetic, 2023. URL `https://arxiv.org/abs/2212.04089`.

J. Kirkpatrick, R. Pascanu, N. Rabinowitz, J. Veness, G. Desjardins, A. A. Rusu, K. Milan, J. Quan, T. Ramalho, A. Grabska-Barwinska, et al. Overcoming catastrophic forgetting in neural networks. *Proceedings of the national academy of sciences*, 114(13):3521–3526, 2017.

J. Krause, M. Stark, J. Deng, and L. Fei-Fei. 3d object representations for fine-grained categorization. In *Proceedings of the IEEE international conference on computer vision workshops*, pages 554–561, 2013.

A. Krizhevsky. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009.

Z. Li and D. Hoiem. Learning without forgetting. *IEEE transactions on pattern analysis and machine intelligence*, 40(12):2935–2947, 2017.

W. Liang, Y. Zhang, Y. Kwon, S. Yeung, and J. Zou. Mind the gap: Understanding the modality gap in multi-modal contrastive representation learning, 2022. URL `https://arxiv.org/abs/2203.02053`.

P. Liu, X. Qiu, and X. Huang. Recurrent neural network for text classification with multi-task learning. *CoRR*, abs/1605.05101, 2016. URL `http://arxiv.org/abs/1605.05101`.

P. Mantini and S. K. Shah. Biclip: Domain canonicalization via structured geometric transformation, 2026. URL `https://arxiv.org/abs/2603.08942`.

M.-E. Nilsback and A. Zisserman. Automated flower classification over a large number of classes. In *2008 Sixth Indian conference on computer vision, graphics & image processing*, pages 722–729. IEEE, 2008.

O. M. Parkhi, A. Vedaldi, A. Zisserman, and C. Jawahar. Cats and dogs. In *2012 IEEE conference on computer vision and pattern recognition*, pages 3498–3505. IEEE, 2012.

J. Pfeiffer, A. Kamath, A. Rücklé, K. Cho, and I. Gurevych. Adapterfusion: Non-destructive task composition for transfer learning. In *Proceedings of the 16th conference of the European chapter of the association for computational linguistics: main volume*, pages 487–503, 2021.

A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever. Learning transferable visual models from natural language supervision, 2021. URL https://arxiv.org/abs/2103.00020.

S. Rebuffi, A. Kolesnikov, and C. H. Lampert. icarl: Incremental classifier and representation learning. *CoRR*, abs/1611.07725, 2016. URL http://arxiv.org/abs/1611.07725.

A. C. Stickland and I. Murray. BERT and pals: Projected attention layers for efficient adaptation in multi-task learning. *CoRR*, abs/1902.02671, 2019. URL http://arxiv.org/abs/1902.02671.

Y.-L. Sung, J. Cho, and M. Bansal. Vl-adapter: Parameter-efficient transfer learning for vision-and-language tasks, 2022. URL https://arxiv.org/abs/2112.06825.

R. Zhang, R. Fang, W. Zhang, P. Gao, K. Li, J. Dai, Y. Qiao, and H. Li. Tip-adapter: Training-free clip-adapter for better vision-language modeling, 2021. URL https://arxiv.org/abs/2111.03930.

K. Zhou, J. Yang, C. C. Loy, and Z. Liu. Learning to prompt for vision-language models. *International Journal of Computer Vision*, 130(9):2337–2348, July 2022a. ISSN 1573-1405. doi: 10.1007/s11263-022-01653-1. URL http://dx.doi.org/10.1007/s11263-022-01653-1.

K. Zhou, J. Yang, C. C. Loy, and Z. Liu. Conditional prompt learning for vision-language models, 2022b. URL https://arxiv.org/abs/2203.05557.

Appendix A: Theoretical Derivations.

A.1 Relation between Orthogonality Error and Spectral Norm

In GeoStack, under the structural constraint, GeoLayers are initialized as Identity matrices and constrained to be upper-triangular. GeoLayer parameters can be represented as $W = \mathbf{I} + \Delta$, where $\mathbf{I}$ is the identity matrix and Δ is an upper-triangular matrix representing a learned perturbation.

From Sec. 2.4, we know that the stability condition will hold if the spectral norm of the perturbation $\|\Delta\|_2$ is small.

We define Orthogonality Error (OE) as the Frobenius norm of deviation from the Identity matrix. This is given by:

$$\mathcal{L}_{ortho} = \|W^\top W - \mathbf{I}\|_F^2$$

Substituting $W = I + \Delta$:

$$\begin{aligned} \|W^\top W - \mathbf{I}\|_F^2 &= \|(\mathbf{I} + \Delta^\top)(\mathbf{I} + \Delta) - \mathbf{I}\|_F^2 \\ &= \|\mathbf{I} + \Delta + \Delta^\top + \Delta^\top\Delta - \mathbf{I}\|_F^2 \\ &= \|\Delta + \Delta^\top + \Delta^\top\Delta\|_F^2 \end{aligned} \tag{8}$$

Under the assumption that W is initialized to the identity and the perturbations are small, the second-order term $\Delta^\top\Delta$ is negligible. Thus, we approximate:

$$\mathcal{L}_{ortho} \approx \|\Delta + \Delta^\top\|_F^2$$

Since $(\Delta + \Delta^\top)$ is symmetric, its entries are $d_{ii} + d_{ii}$ on the diagonal and u_{ij} or u_{ji} on the off-diagonals. One can see that:

$$\|\Delta + \Delta^\top\|_F^2 = 2\|\Delta\|_F^2 + 2\sum_i d_{ii}^2$$

Thus, we establish a lower bound relative to the perturbation:

$$\|\Delta + \Delta^\top\|_F^2 \geq 2\|\Delta\|_F^2$$

$$\implies \mathcal{L}_{ortho} \gtrsim 2\|\Delta\|_F^2$$

Finally, since the Frobenius norm upper bounds the spectral norm ($\|\Delta\|_2 \leq \|\Delta\|_F$), we have:

$$2\|\Delta\|_2^2 \leq 2\|\Delta\|_F^2 \lesssim \mathcal{L}_{ortho}$$

$$\implies \|\Delta\|_2 \lesssim \sqrt{\frac{\mathcal{L}_{ortho}}{2}}$$

Consequently, by minimizing the Orthogonality Error $\mathcal{L}_{ortho}$, the optimization process suppresses the growth of the spectral norm of the perturbation $\|\Delta\|_2$. Ultimately, this fulfills the stability condition required for GeoStack, allowing multiple domain experts to be integrated without the destructive interference from one another.

Appendix B: Scaling Behavior.

B.1 Dual Stack Analysis

In this section, we present the performance of GeoStack for MDA using dual-stack compositions. We assume ImageNet as the foundational expertise and quantify the performance of GeoStack under an additional cross-domain composition. We compare the standard BiCLIP adapter (baseline) against the GeoStack framework. Stability is measured via the Orthogonality Error (OE) and the retention of foundational knowledge on ImageNet.

Table 4: Dual-Stack Performance Comparison. Brackets $[\cdot]$ denote the Orthogonality Error (OE). Zero-shot values represent the performance of the frozen CLIP backbone.

Stack Sequence	Dataset	Zero-Shot	BiCLIP (Baseline)	GeoStack (Ours)
Sequence A $i \to c$	ImageNet (i)	66.6%	69.1% [0.011]	**69.6% [0.006]**
	Caltech-101 (c)	90.0%	93.5% [0.011]	**93.9% [0.006]**
	Average	78.3%	81.3%	**81.8%**
Sequence B $i \to fo$	ImageNet (i)	66.6%	69.6% [0.011]	**70.2% [0.007]**
	Food-101 (fo)	88.7%	89.6% [0.011]	**90.0% [0.007]**
	Average	77.7%	79.6%	**80.1%**
Sequence C $i \to e$	ImageNet (i)	66.6%	67.6% [0.023]	**69.6% [0.007]**
	EuroSAT (e)	47.5%	**85.7% [0.023]**	84.7% [0.007]
	Average	57.1%	76.7%	**77.2%**
Sequence D $i \to d$	ImageNet (i)	66.6%	59.5% [0.050]	**66.3% [0.009]**
	DTD (d)	42.8%	**69.7% [0.050]**	68.4% [0.009]
	Average	54.7%	64.6%	**67.4%**
Sequence E $i \to f$	ImageNet (i)	66.6%	65.8% [0.026]	**67.9% [0.010]**
	Flowers (f)	71.0%	**94.2% [0.026]**	92.7% [0.010]
	Average	68.8%	80.0%	**80.3%**

The dual-stack experiments (Table 4) reveal a clear correlation between Orthogonality Error (OE) and foundational knowledge retention. Across all five sequences, GeoStack consistently maintains higher ImageNet accuracy compared to the BiCLIP baseline. This is most evident in Sequence D (DTD), where the baseline's OE explodes to 0.050, resulting in a catastrophic 7.1% drop in ImageNet performance compared to the zero-shot baseline. In contrast, GeoStack restricts the OE to 0.009, preserving the foundational knowledge within 0.4% of its original state.

B.2 Triple Stack Analysis

In this section, we present the performance of GeoStack for MDA using three triple-stack compositions simialr to the quad-stack compositions: Easy ($i \to c \to fo$), Moderate ($i \to fo \to e$), and Hard ($i \to d \to f$).

Table 5: Triple-Stack Performance Comparison. The results highlight how Orthogonality Error (OE) accumulates across three-expert compositions.

Stack Sequence	Dataset	Zero-Shot	BiCLIP (Baseline)	GeoStack (Ours)
Easy Stack $i \to c \to fo$	ImageNet (i)	66.6%	69.1% [0.013]	**69.7% [0.009]**
	Caltech-101 (c)	90.0%	**93.4% [0.013]**	93.2% [0.009]
	Food-101 (fo)	88.7%	89.4% [0.013]	**89.7% [0.009]**
	Average	81.8%	84.0%	**84.2%**
Moderate Stack $i \to fo \to e$	ImageNet (i)	66.6%	67.7% [0.022]	**69.7% [0.009]**
	Food-101 (fo)	88.7%	88.3% [0.022]	**89.6% [0.009]**
	EuroSAT (e)	47.5%	**84.5% [0.022]**	83.9% [0.009]
	Average	67.6%	80.2%	**81.1%**
Hard Stack $i \to d \to f$	ImageNet (i)	66.6%	55.2% [0.062]	**63.6% [0.013]**
	DTD (d)	42.8%	**69.7% [0.062]**	67.2% [0.013]
	Flowers (f)	71.0%	**88.8% [0.062]**	87.2% [0.013]
	Average	59.8%	71.2%	**72.7%**

BiCLIP degrades under a triple stack composition and shows catastrophic forgetting. ImageNet accuracy drops to 55.2% in the hard stack. GeoStack maintains an OE of only 0.013, keeping

ImageNet accuracy at 63.6%. While there is a slight drop from zero-shot, it is 8.4% higher than the baseline.

B.3 Beyond Quad Stack

To determine the upper limits of manifold stability, we evaluate on two additional datasets, Stanford-Cars [Krause et al., 2013] (s), and Oxford-Pets [Parkhi et al., 2012] (o). We evaluate the performance on Hexa-stack composition ($i \to o \to f \to s \to e \to d$). This sequence represents a significant accumulation of disparate domain knowledge, including fine-grained animals, flowers, vehicles, satellite imagery, and textures.

Table 6: Hexa-Stack Performance Comparison. OE denotes the final cumulative Orthogonality Error after all six experts are folded into the backbone.

Dataset	**Zero-Shot**	**BiCLIP (Baseline)**	**GeoStack (Ours)**
ImageNet (i)	66.6%	39.7%	**62.2%**
Oxford-Pet (o)	89.0%	72.7%	**86.3%**
Flowers-102 (f)	71.0%	71.0%	**86.8%**
Stanford-Cars (s)	66.3%	**65.2%**	60.2%
EuroSAT (e)	47.5%	72.4%	**81.7%**
DTD (d)	42.8%	62.8%	**63.2%**
Average Acc.	63.9%	64.0%	**73.4%**
Final OE $\downarrow$	—	0.1359	**0.0142**

The results in Table 6 demonstrate the results for Hexa-stack composition. The BiCLIP baseline collapses with an accumulated OE (0.1359). This is most visible in the Oxford-Pet accuracy, which drops below zero-shot for BiCLIP (72.7%) but remains at 86.3% for GeoStack. Notably, GeoStack achieves a +9.4% lead in Average Accuracy across all tasks. While BiCLIP shows higher plasticity on a single task (Stanford Cars), it does so by sacrificing the integrity of all other experts. GeoStack's ability to maintain an OE of 0.0142—ten times lower than the baseline—validates its use as a scalable architecture for knowledge composition.

Appendix C: Stacking Metric.

C.1 Orthogonality Error as a Metric for Stackability

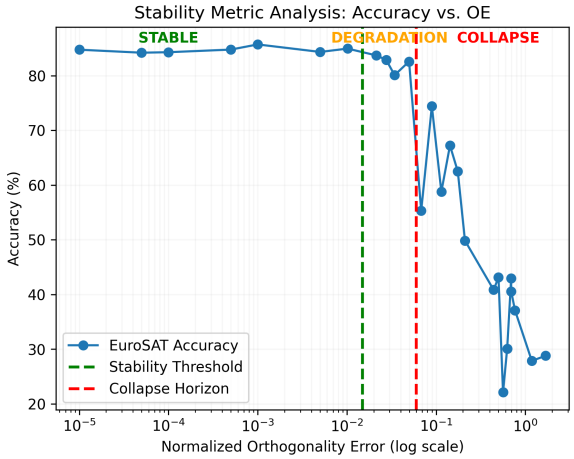

Figure 5: **Stability Stress Test Analysis.** EuroSAT accuracy as a function of simulated Orthogonality Error (OE).

In Sec. 2.5, we defined the stackability metric as the normalized Orthogonality Error (OE):

$$\mathcal{S}(W)^2 = \frac{1}{d^2}\|W^\top W - \mathbf{I}\|_F^2$$

Where d is the model dimensions. Scaling by $1/d^2$ ensures that the error remains comparable even if the model's embedding dimension d changes.

Here, we provide empirical results to calibrate this metric and identify the specific ranges that determine compatibility for weight folding. We conducted experiments by injecting synthetic experts with controlled OE. We synthesize matrices to match a target orthogonality error $\gamma \in [10^{-5}, 1.7]$. We folded these experts into a frozen backbone and evaluated the accuracy on the EuroSAT dataset.

The results shown in Fig. 5 reveal three distinct ranges:

- **The Stable Plateau ($\mathcal{S} < 0.015$):** Accuracy remains consistent in this range. The perturbation is small enough that the latent decision margins M_a can absorb the interference without displacing previous knowledge.
- **The Graceful Degradation Zone ($0.015 \leq \mathcal{S} < 0.06$):** The model retains most task-specific knowledge, the decision boundaries shift enough to cause a gradual decline in accuracy ($1\% \rightarrow 5\%$ drop).
- **The Catastrophic Forgetting Horizon ($\mathcal{S} \geq 0.06$):** A threshold is crossed where the foundational knowledge is corrupted. This leads to a collapse of class separation, resulting in the rapid degradation in performance.

Appendix D: Broader Impacts.

D.1 Positive Societal Impact

The primary positive impact of GeoStack is environmental sustainability. By enabling the folding of multiple domain experts into a single $O(1)$ inference operation, our method significantly reduces the computational energy and memory overhead required for multi-task deployments.

D.2 Negative Societal Impact & Limitations

As a foundational method for adapting pre-trained models like CLIP, GeoStack inherits the inherent biases and fairness issues present in the backbone architecture. GeoStack does not inherently audit or filter the semantic content of the experts being stacked. Consequently, an intentional misuse of the framework could involve the sequential stacking of harmful or biased experts.

D.3 Mitigation

Because GeoStack transformations are represented as transparent $d \times d$ linear layers rather than deep black-box adapters, they are more amenable to weight-space auditing.

Appendix E: Licenses.

E.1 Assets

Table 7: Asset Documentation: Versions, Licenses, and Access URLs.

Asset	Version	License	Access / URL
OpenCLIP	ViT-B/16	MIT	GitHub
ImageNet	2012 (1k)	Custom	image-net.org
CIFAR-100	PyTorch/Torchvision	MIT	cs.toronto.edu
Caltech-101	PyTorch/Torchvision	CC BY 4.0	caltech.edu
Flowers-102	PyTorch/Torchvision	Custom	ox.ac.uk
Food-101	PyTorch/Torchvision	Custom	ethz.ch
DTD	v1.0	CC BY-SA 4.0	ox.ac.uk
EuroSAT	RGB Version	MIT	GitHub
Stanford Cars	2013	Custom	stanford.edu
Oxford-Pets	v1.1	CC BY-SA 4.0	ox.ac.uk

Appendix F: Compute Resources.

F.1 Computational Resources and Efficiency

All experiments were conducted on a workstation equipped with an **NVIDIA GeForce RTX 2080 Ti (11GB VRAM)**.

Memory Usage: During the training of a single GeoLayer adapter with a **ViT-B/16** backbone, peak memory usage was approximately **8.2 GB**. After weight folding, the inference memory footprint remains identical to that of the vanilla CLIP-ViT-B/16 projection head.

Training Time: Each domain-specific expert was trained for 10 to 30 epochs, with an average execution time of **25 minutes** per expert on a single NVIDIA RTX 2080 Ti.